A/C HIGH PRESSURE SWITCH (Continued)

the refrigerant system pressure exceeds 3100 - 3375 kPa (450 - 490 psi).

The fitting for the A/C high pressure switch on the A/C discharge line is equipped with an O-ring seal and contains a Schrader-type valve, which allows the switch to be serviced without discharging the refrigerant system.

OPERATION

The A/C high pressure switch is electrically connected in series with the A/C low pressure switch between ground and the powertrain control module (PCM). The contacts in the A/C high pressure switch open and close causing the PCM to turn the A/C compressor clutch on and off. This prevents A/C compressor operation when the discharge line pressure approaches high levels, and also reduces electrical surging from A/C clutch engagement.

The A/C high pressure switch contacts open when the A/C discharge line pressure rises above 3100 to 3375 kPa (450 to 490 psi). The A/C high pressure switch contacts close when the A/C discharge line pressure drops to 1860 to 2275 kPa (270 to 330 psi).

The A/C high pressure switch is factory-calibrated and cannot be adjusted or repaired and, if faulty or damaged, it must be replaced.

DIAGNOSIS AND TESTING

A/C HIGH PRESSURE SWITCH

Before performing diagnosis of the A/C high pressure switch, verify that the refrigerant system has the correct refrigerant charge (Refer to 24 - HEATING & AIR CONDITIONING/PLUMBING - STANDARD PROCEDURE - REFRIGERANT SYSTEM CHARGE)

For circuit descriptions and diagrams, refer to the appropriate wiring information. The wiring information includes wiring diagrams, proper wire and connector repair procedures, further details on wire harness routing and retention, as well as pin-out and location views for the various wire harness connectors, splices and grounds.

(1) Disconnect and isolate the negative battery cable.

(2) Disconnect the wire harness connector from the A/C high pressure switch.

(3) Check for continuity between both terminals of the A/C high pressure switch. There should be continuity. If there is not continuity, replace the faulty A/C high pressure switch. If there is continuity, test the A/C high pressure switch sense circuit (Refer to 8 - ELECTRICAL/WIRING DIAGRAM INFORMATION - DIAGNOSIS AND TESTING). Repair as required.

REMOVAL

NOTE: It is not necessary to discharge the refrigerant system to replace the A/C high pressure switch.

(1) Disconnect and isolate the negative battery cable.

(2) Disconnect the wire harness connector from the A/C high pressure switch (Fig. 18).

(3) Remove the A/C high pressure switch from the A/C discharge line and remove and discard the O-ring seal.

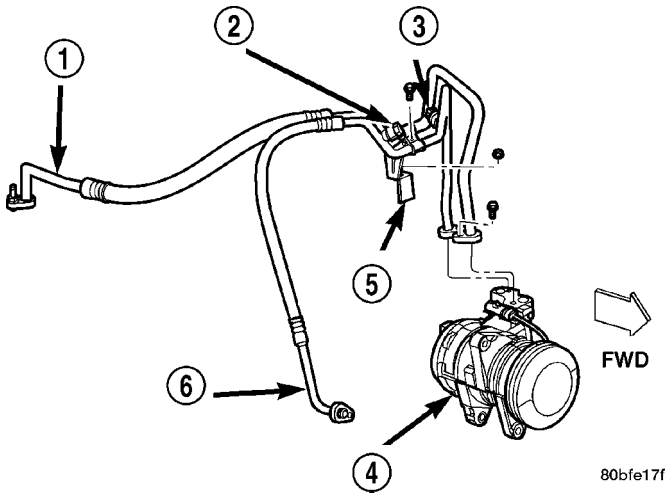

Fig. 18 A/C High Pressure Switch - Typical

1 - A/C SUCTION LINE
2 - SERVICE PORT
3 - A/C HIGH PRESSURE SWITCH
4 - A/C COMPRESSOR
5 - MOUNTING BRACKET
6 - A/C DISCHARGE LINE

INSTALLATION

NOTE: Use only the specified O-ring as it is made of special material for R-134a. Use only refrigerant oil of the type required for the A/C compressor.

(1) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the A/C high pressure switch fitting.

(2) Install the A/C high pressure switch onto the A/C discharge line. Hand-tightened the switch securely.

(3) Connect the wire harness connector to the A/C high pressure switch.

(4) Reconnect the negative battery cable.